

Analytical workload	Workload-critical information in R2	Analytical consequence (R2)	Query-only performance vs R0
W1 Acute bronchitis cohort	Clinical code + patient association PRESERVED	Feasible Exact fidelity, F1 = 1.000	3.94× faster than R0 (query-only)
W2 Condition-medication cohort	Condition + medication codes + patient associations PRESERVED	Feasible Exact fidelity, F1 = 1.000	5.16× faster than R0 (query-only)
W3 Relationship-aware temporal workload	MedRequest→Condition relationship + authoredOn + onsetDateTime PRESERVED	Feasible Exact fidelity, F1 = 1.000	1.21× faster than R0 (query-only)
W4 Encounter duration > 24 h	Required: period.start + period.end period.start: PRESERVED period.end: REMOVED	Not feasible Fidelity: N/A	N/A not measurable (query infeasible)

W3 — only 50.56% of temporal values are retained overall in R2, but the specific timestamps W3 needs (authoredOn, onsetDateTime) survive → fidelity remains exact.

W4 — temporal information is still partially present (period.start), but the one workload-critical timestamp it needs (period.end) is removed → the workload becomes impossible.

Analytical impact depends on workload-critical information, not on overall information retention alone.